Equation Sheet: Numerical solutions to 2D parabolic and elliptic PDEs

1 PDEs and Boundary Conditions

$$\kappa(x,y) = 0.02\, e^{-(x-0.7)^2/0.09 \;-\; (y-0.6)^2/0.25}$$

$$\alpha\left(\frac{\partial^2 \tilde{u}}{\partial x^2} + \frac{\partial^2 \tilde{u}}{\partial y^2}\right) = -\kappa \qquad \text{(Poisson)}$$

$$\frac{\partial u}{\partial t} = \alpha\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right) + \kappa, \qquad u(x,y,0) = 0 \qquad \text{(Heat)}$$

$$u = 0.5 - 0.5\cos(2\pi y) \qquad x = 0 \text{ (West, Dirichlet)} \tag{1}$$
$$\partial_x u = 0 \qquad x = 1 \text{ (East, Neumann)} \tag{2}$$
$$\partial_y u = -0.3 \qquad y = 0 \text{ (South, Neumann)} \tag{3}$$
$$u = 0.5 + 0.5\sin\!\left(4\pi x - \tfrac{\pi}{2}\right) \qquad y = 1 \text{ (North, Dirichlet)} \tag{4}$$

2 Grid and Index Map

$$\Delta x = \frac{1}{N_x + 1}, \qquad \Delta y = \frac{1}{N_y + 1}$$

$$x_i = \frac{i}{N_x + 1}, \quad i = 1, \ldots, N_x \qquad y_j = \frac{j}{N_y + 1}, \quad j = 1, \ldots, N_y$$

$$k = \mathtt{iQ}(i,j) = (j-1)\,N_x + i$$

3 Discrete Laplacian Operator Split

$$\alpha\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right) \approx \alpha(\mathbf{LQ} + \mathbf{L}_{\mathrm{BC}})$$

Interior stencil $(2 \le i \le N_x - 1,\ 2 \le j \le N_y - 1)$

$$(\mathbf{LQ})_{i,j} = \frac{Q_{i-1,j} - 2Q_{i,j} + Q_{i+1,j}}{\Delta x^2} + \frac{Q_{i,j-1} - 2Q_{i,j} + Q_{i,j+1}}{\Delta y^2}$$

Boundary stencils

The diagonal coefficient on $Q_{i,j}$ changes at boundary nodes. All present off-diagonal neighbours retain coefficient $+1/h^2$.

Region	x-diagonal on $Q_{i,j}$	y-diagonal on $Q_{i,j}$
Interior	$-2/\Delta x^2$	$-2/\Delta y^2$
West ($i = 1$, Dirichlet)	$-2/\Delta x^2$	$-2/\Delta y^2$
East ($i = N_x$, Neumann)	$-1/\Delta x^2$	$-2/\Delta y^2$
South ($j = 1$, Neumann)	$-2/\Delta x^2$	$-1/\Delta y^2$
North ($j = N_y$, Dirichlet)	$-2/\Delta x^2$	$-2/\Delta y^2$
Corner BL ($i = 1, j = 1$)	$-2/\Delta x^2$	$-1/\Delta y^2$
Corner BR ($i = N_x, j = 1$)	$-1/\Delta x^2$	$-1/\Delta y^2$
Corner TL ($i = 1, j = N_y$)	$-2/\Delta x^2$	$-2/\Delta y^2$
Corner TR ($i = N_x, j = N_y$)	$-1/\Delta x^2$	$-2/\Delta y^2$

Ghost-point derivations

East ($\partial_x u = 0$):

$$Q_{N_x+1,j} = Q_{N_x-1,j} \implies \frac{Q_{N_x-1,j} - 2Q_{N_x,j} + Q_{N_x+1,j}}{\Delta x^2} = \frac{Q_{N_x-1,j} - Q_{N_x,j}}{\Delta x^2}$$

South ($\partial_y u = -0.3$):

$$Q_{i,0} = Q_{i,2} - 2\Delta y\,(0.3) \implies \frac{Q_{i,0} - 2Q_{i,1} + Q_{i,2}}{\Delta y^2} = \frac{-Q_{i,1} + Q_{i,2}}{\Delta y^2} + \frac{2(-0.3)}{\Delta y}$$

4 BC Correction Vector $\mathbf{L}_{\mathbf{BC}}$

$$(\mathbf{L}_{\mathrm{BC}})_{1,j} \mathrel{+}= \frac{\mathrm{BCL\_D}(j)}{\Delta x^2} \qquad \text{West Dirichlet}$$

$$(\mathbf{L}_{\mathrm{BC}})_{i,N_y} \mathrel{+}= \frac{\mathrm{BCT\_D}(i)}{\Delta y^2} \qquad \text{North Dirichlet}$$

$$(\mathbf{L}_{\mathrm{BC}})_{N_x,j} \mathrel{+}= \frac{\mathrm{BCR\_N}(j)}{\Delta x} \qquad \text{East Neumann}$$

$$(\mathbf{L}_{\mathrm{BC}})_{i,1} \mathrel{+}= \frac{-\mathrm{BCB\_N}(i)}{\Delta y} \qquad \text{South Neumann}$$

Corners accumulate both adjacent contributions, e.g.:

$$(\mathbf{L}_{\mathrm{BC}})_{1,1} = \frac{\mathrm{BCL\_D}(1)}{\Delta x^2} + \frac{-\mathrm{BCB\_N}(1)}{\Delta y} \qquad \text{(Bottom-Left, D+N)}$$

5 Poisson Linear System

$$\underbrace{\alpha\,\mathbf{L}}_{\mathbf{A}_{\mathrm{EL}}}\,\mathbf{Q} = -\boldsymbol{\kappa} - \alpha\,\mathbf{L}_{\mathrm{BC}}$$

6 Conjugate Gradient Algorithm

Initialise: $\mathbf{r}_0 = \mathbf{b} - \mathbf{A}\mathbf{x}_0,\ \mathbf{d}_0 = \mathbf{r}_0,\ \delta_{\mathrm{new}} = \mathbf{r}_0^T\mathbf{r}_0,\ \delta_0 = \delta_{\mathrm{new}}$.

$$\begin{aligned}
\mathbf{q} &= \mathbf{A}\,\mathbf{d} \\
\alpha &= \delta_{\mathrm{new}} \,/\, (\mathbf{d}^T\mathbf{q}) \\
\mathbf{x} &\leftarrow \mathbf{x} + \alpha\,\mathbf{d} \\
\mathbf{r} &\leftarrow \mathbf{r} - \alpha\,\mathbf{q} \\
\delta_{\mathrm{new}} &= \mathbf{r}^T\mathbf{r} \\
\beta &= \delta_{\mathrm{new}} \,/\, \delta_{\mathrm{old}} \\
\mathbf{d} &\leftarrow \mathbf{r} + \beta\,\mathbf{d}
\end{aligned}$$

Stop when $\delta_{\mathrm{new}} \le \varepsilon^2\,\delta_0$.

7 Crank–Nicolson Time Stepping

$$\frac{\mathbf{Q}^{n+1} - \mathbf{Q}^n}{\Delta t} = \frac{\alpha}{2}\big[\mathbf{L}\mathbf{Q}^n + \mathbf{L}_{\mathrm{BC}} + \mathbf{L}\mathbf{Q}^{n+1} + \mathbf{L}_{\mathrm{BC}}\big] + \boldsymbol{\kappa}$$

$$\underbrace{\left(\mathbf{I} - \frac{\alpha\Delta t}{2}\mathbf{L}\right)}_{\mathbf{A}_{\mathrm{CN}}}\mathbf{Q}^{n+1} = \frac{\alpha\Delta t}{2}(\mathbf{L}\mathbf{Q}^n + 2\,\mathbf{L}_{\mathrm{BC}}) + \Delta t\,\boldsymbol{\kappa} + \mathbf{Q}^n$$

$$\mu = 1 + \frac{\alpha\Delta t}{2}\,\lambda \ \ge\ 1 \qquad \text{(eigenvalues of } \mathbf{A}_{\mathrm{CN}},\ \lambda \ge 0 \text{ from } -\mathbf{L})$$

$$\Delta t_{\max}^{\mathrm{explicit}} = \frac{\Delta x^2}{4\alpha} \qquad \frac{\|\mathbf{Q}^n - \mathbf{Q}_{\mathrm{ss}}\|_2}{\|\mathbf{Q}_{\mathrm{ss}}\|_2} \le 10^{-4}$$

8 Neumann Boundary Recovery

$$u_{\mathrm{East}}(y_j) = Q_{N_x,j} + \Delta x \cdot \mathrm{BCR\_N}(j)$$

$$u_{\mathrm{South}}(x_i) = Q_{i,1} - \Delta y \cdot \mathrm{BCB\_N}(i)$$